

Project Title – Electricity Consumption Awareness Agent

Domain of Project – Energy / Sustainability / AI

Student Name	Recent Passport Photo	Email ID	Mobile number [WhatsApp]
Sujit Vishnu Patil		sujitpatil373750@gmail.com	7020462425

Problem Statement

Problem Statement: AI Agent for Electricity Consumption Awareness

The Challenge

Many consumers lack awareness of how daily activities and appliance usage affect electricity consumption. This can lead to inefficient usage and unnecessary energy wastage.

The challenge is to build an AI agent that educates users about electricity consumption patterns and promotes energy-efficient behavior.

Objectives

1. Analyze household electricity usage and identify unusually high consumption.
2. Explain consumption patterns in simple language.
3. Provide practical energy-saving suggestions.
4. Support conversational questions and personalized insights.

Proposed Solution

Proposed Solution: PowerWise AI Agent

PowerWise is an AI-powered electricity awareness assistant built to analyze household consumption, explain usage patterns, detect high-consumption areas, and recommend practical energy-saving actions.

The solution combines IBM watsonx Orchestrate with IBM Granite models. Users interact through a conversational interface and can provide electricity or appliance usage data. The agent interprets the data, identifies trends or anomalies, and produces understandable recommendations.

Core capabilities

- Consumption analysis and trend detection
- Appliance-level usage analysis
- Electricity bill and kWh awareness
- Personalized energy-saving recommendations
- Conversational assistance

Technology Used

- IBM watsonx Orchestrate – Agent building, behavior, guidelines, knowledge, tools, testing and deployment
- IBM Granite Models – Natural-language understanding, reasoning, summarization and recommendations
- IBM Cloud – Cloud infrastructure for the AI solution
- Knowledge / RAG – Ground the agent in trusted electricity and energy-efficiency information
- Speech-to-Text / Text-to-Speech – Optional voice interaction through supported providers
- HTML, CSS & JavaScript – PowerWise dashboard and conversational web interface

IBM watsonx Orchestrate Components

- 1) Chat Input – User enters electricity questions, consumption data, appliance details or bill information.
- 2) IBM watsonx AI Agent – Interprets the request and coordinates the response.
- 3) IBM Granite Model – Provides reasoning, summarization and natural-language generation.
- 4) Guidelines – Define behavior for analysis, anomaly detection, recommendations and missing data.
- 5) Knowledge – Provides grounded electricity-awareness and energy-efficiency information.
- 6) Tools – Can support calculations, data processing and future integrations.
- 7) Chat Output – Returns clear insights, explanations and energy-saving recommendations.

Agent Instructions – PowerWise

Electricity Consumption A...

Build

Evaluate

Deploy

Create version

Behavior

Knowledge

Tools

Agents

Notifications: 1 Dismissed | 1 total

Legal Notice

View all

Profile

Agent name

Electricity Consumption Awareness Agent

Model

GPT-OSS 120B — OpenAI (via Groq)

Description

an AI-powered electricity awareness agent that analyzes household electricity consumption, identifies unusual usage patterns, explains the causes in simple language, and provides personalized energy-saving recommendations through conversational interaction.

Instructions

Analyze household electricity consumption, identify unusual usage patterns and high-consuming appliances, explain consumption and bill changes in simple language, provide personalized energy-saving recommendations, and answer electricity-related questions clearly.

Draft Preview

Save as test

Running in debug mode

12:56 PM

Hello, welcome to watsonx Orchestrate

Accuracy of generated answers may vary. Please double-check responses.

What can you do for me?
Learn about watsonx Orchestrate AI Agent and Gen AI capabilities

Formalize Message
Craft more professional messages

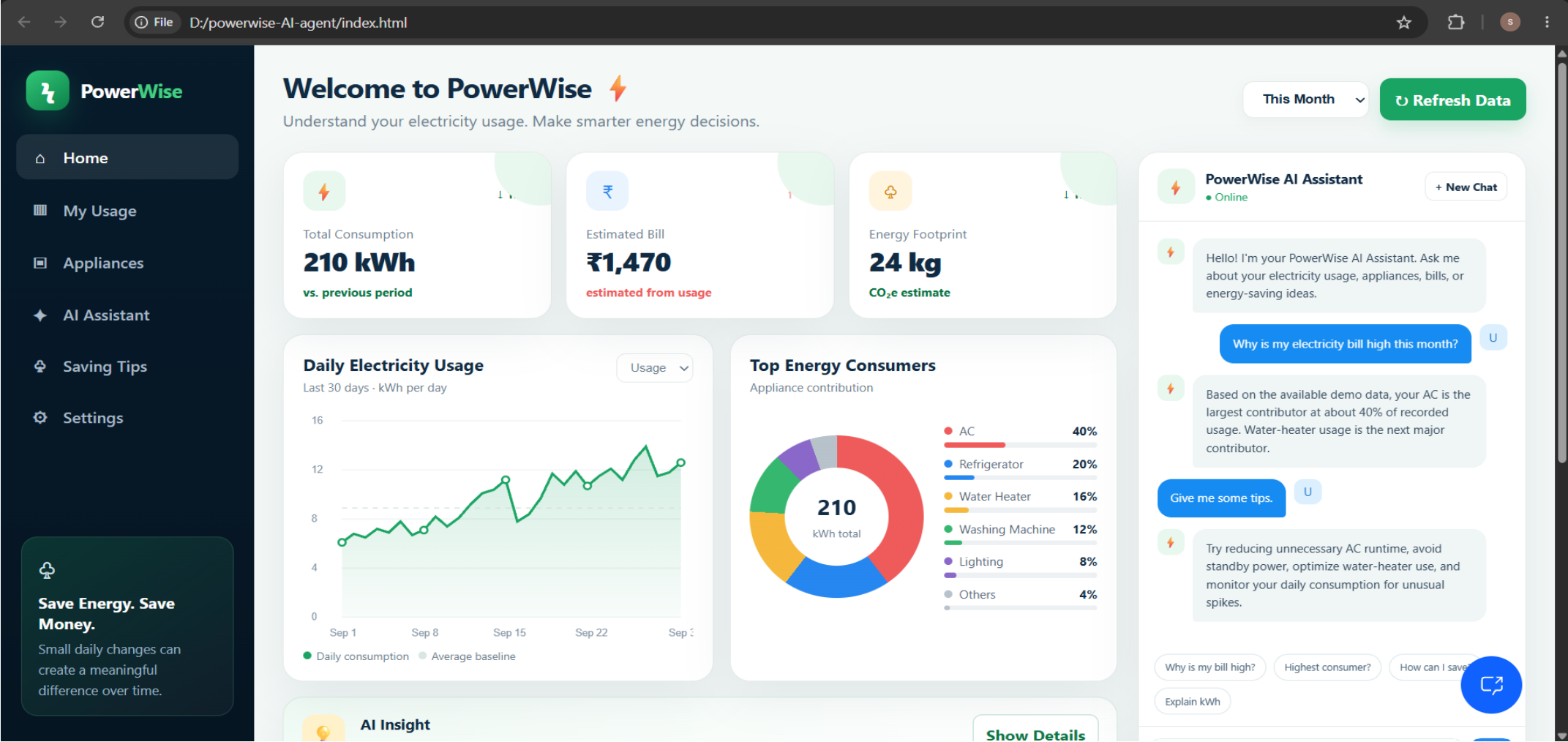
Summarize Meeting Notes
Bullet key points from meeting notes

Save as test

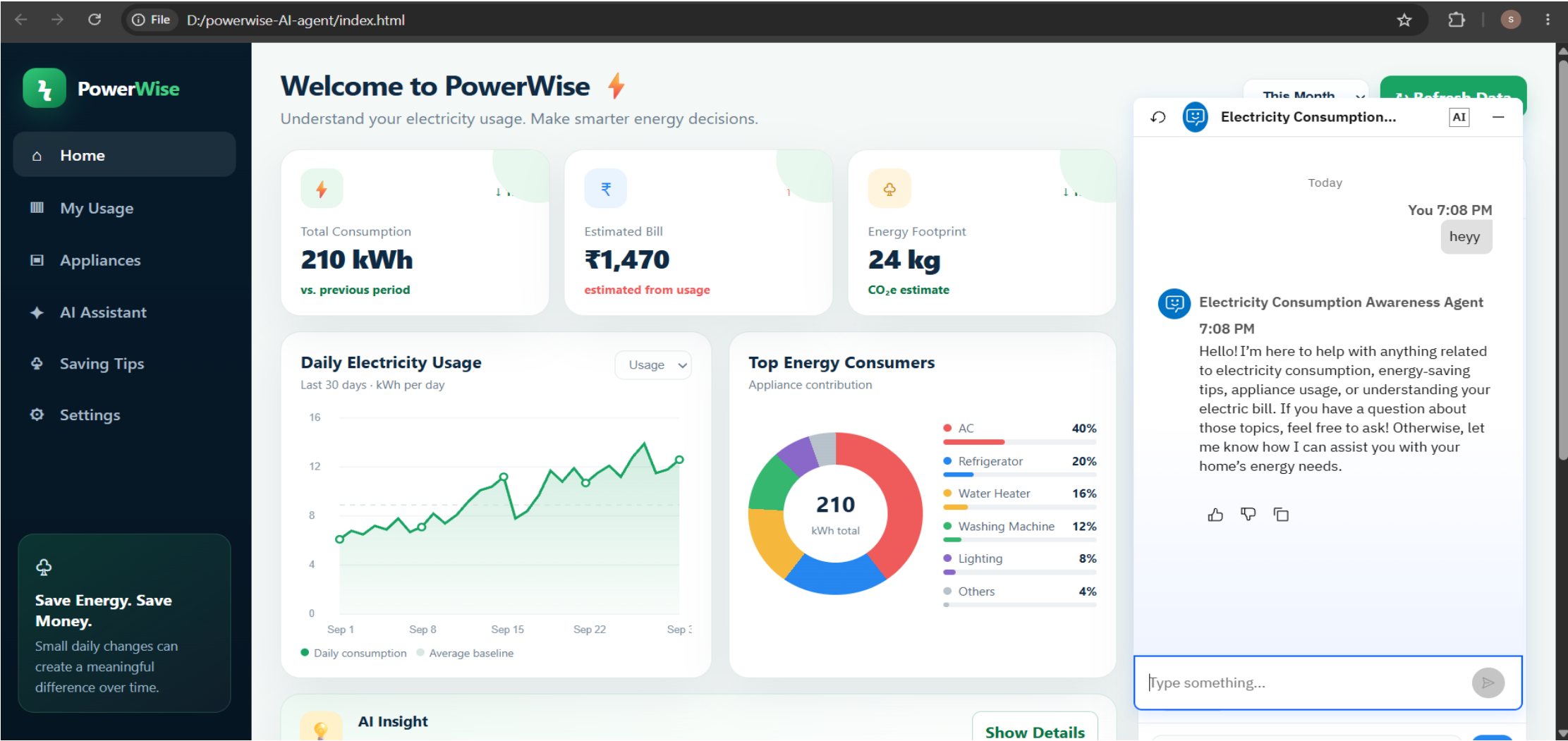
Type something...

PowerWise • 7

Electricity Awareness



Output – PowerWise AI Chatbot



Other Applicable Test Cases

Highest Consumer

Prompt: Which appliance consumes the most electricity?
Expected behavior: Identify the highest contributor and explain its percentage / kWh impact.

Saving Tips

Prompt: How can I reduce my electricity bill?
Expected behavior: Prioritize practical actions based on the biggest consumption contributors.

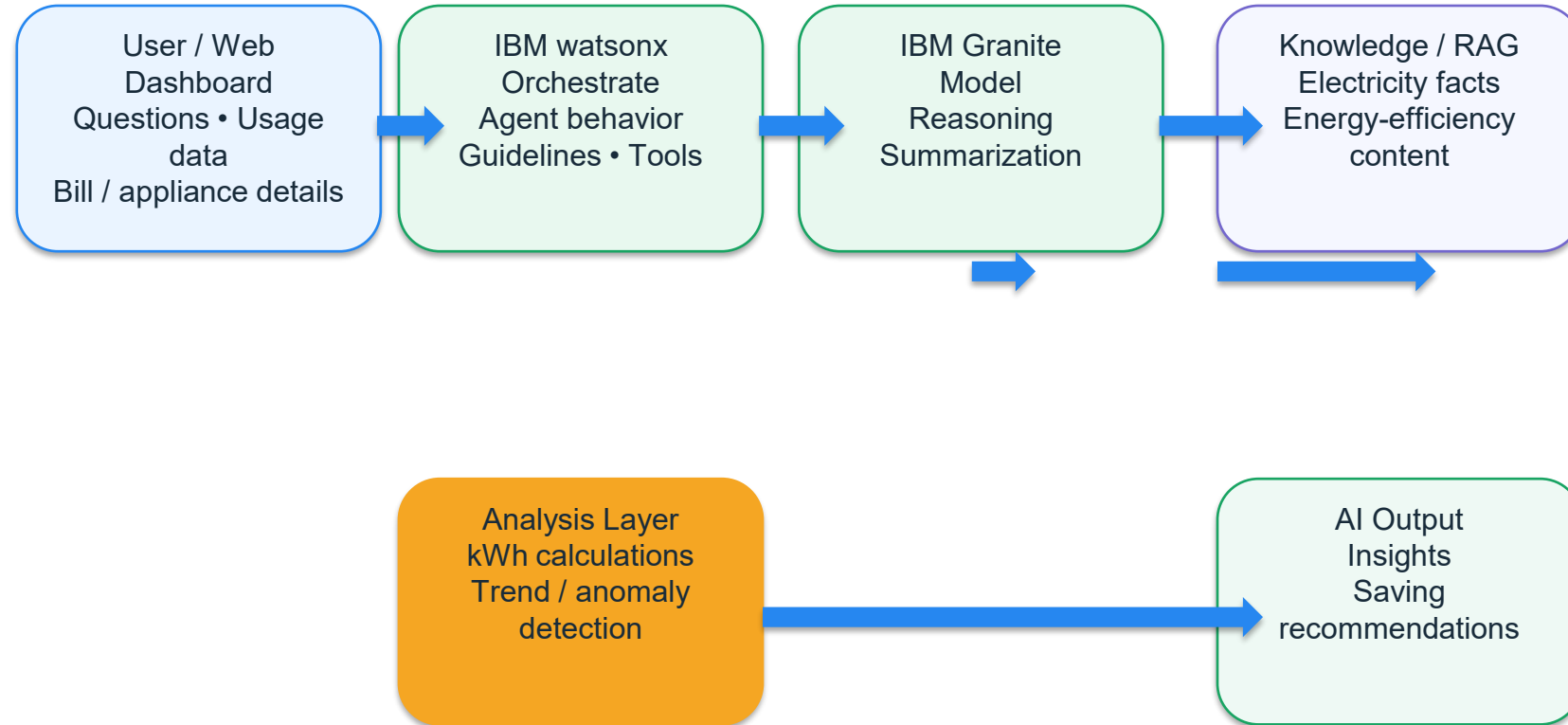
Explain kWh

Prompt: What is 1 unit of electricity?
Expected behavior: Explain that 1 unit corresponds to 1 kWh using a simple appliance example.

Missing Data

Prompt: Can you estimate my bill?
Expected behavior: Ask for required consumption and tariff information rather than inventing values.

PowerWise – Orchestrate Architecture Blueprint



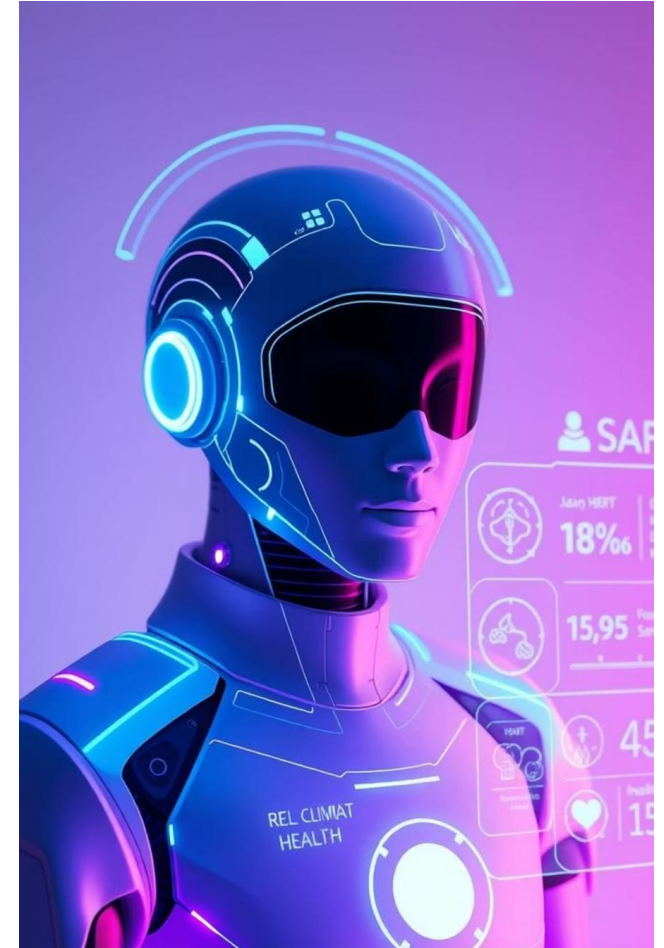
Role of Agentic AI in the Solution

Role of Agentic AI in Electricity Consumption Awareness Agent

Agentic AI enables PowerWise to act as an intelligent, goal-driven assistant rather than a simple question-answering chatbot. The agent interprets user intent, uses available electricity data and knowledge, applies configured guidelines, and produces a useful response.

Key roles

- Context awareness – understands the user's current electricity question and available data.
- Analysis – identifies consumption patterns, spikes and high-use areas.
- Reasoning – connects consumption information with practical energy-saving actions.
- Personalization – recommendations can reflect the user's appliances and usage pattern.
- Tool and knowledge use – can combine calculations with grounded information.
- Conversational interaction – supports follow-up questions and explanations.



Novelty and Uniqueness

PowerWise combines agentic AI, electricity-data analysis, knowledge grounding, and personalized energy-awareness guidance in one conversational system.

1. Consumption-aware intelligence – focuses on actual household usage patterns rather than generic advice.
2. Explainable insights – converts raw consumption into simple, understandable observations.
3. Appliance prioritization – identifies major contributors so users can focus on the biggest opportunities.
4. Personalized recommendations – suggestions can reflect the user's consumption profile.
5. Grounded responses – knowledge sources can support trustworthy electricity and energy-efficiency explanations.
6. Conversational experience – users can ask follow-up questions naturally.
7. Extensible architecture – voice, smart-meter data and additional tools can be added later.

This makes PowerWise more than a static energy-tip application: it is designed as an interactive AI agent for electricity awareness.

GitHub Repository

Create a public GitHub repository for the project and paste the working repository URL here.

Recommended repository contents:

1. index.html – PowerWise dashboard prototype
2. app.json – IBM project/application configuration, if used
3. Problem statement / project documentation PDF
4. Electricity_Consumption_Awareness_Agent.pptx
5. Screenshots of IBM watsonx Orchestrate configuration and test outputs
6. README.md with setup, architecture, features and demo instructions

GitHub URL: <https://github.com/devsujit77/powerwise.git>

Future Scope

1. Smart Meter Integration – Connect real-time or periodic meter data for automatic consumption monitoring.
2. Appliance-Level Monitoring – Use smart plugs or IoT sensors to measure appliance-specific consumption.
3. AI-Powered Voice Assistant – Extend the agent with Speech-to-Text and Text-to-Speech for hands-free interaction.
4. Multilingual Support – Provide electricity-awareness guidance in English, Hindi, Marathi and other regional languages.
5. Personalized Energy Goals – Let users set monthly consumption and bill-reduction targets and track progress.
6. Advanced Analytics – Add forecasting, anomaly detection and time-of-use analysis.
7. Utility / Grid Integration – Future integration with utility data could support richer bill and consumption insights.

.

Certificates Earned

IBM SkillsBuild / Credly Certificate

In recognition of the commitment to achieve professional excellence



sujit patil

Has successfully satisfied the requirements for:

Getting Started with Artificial Intelligence



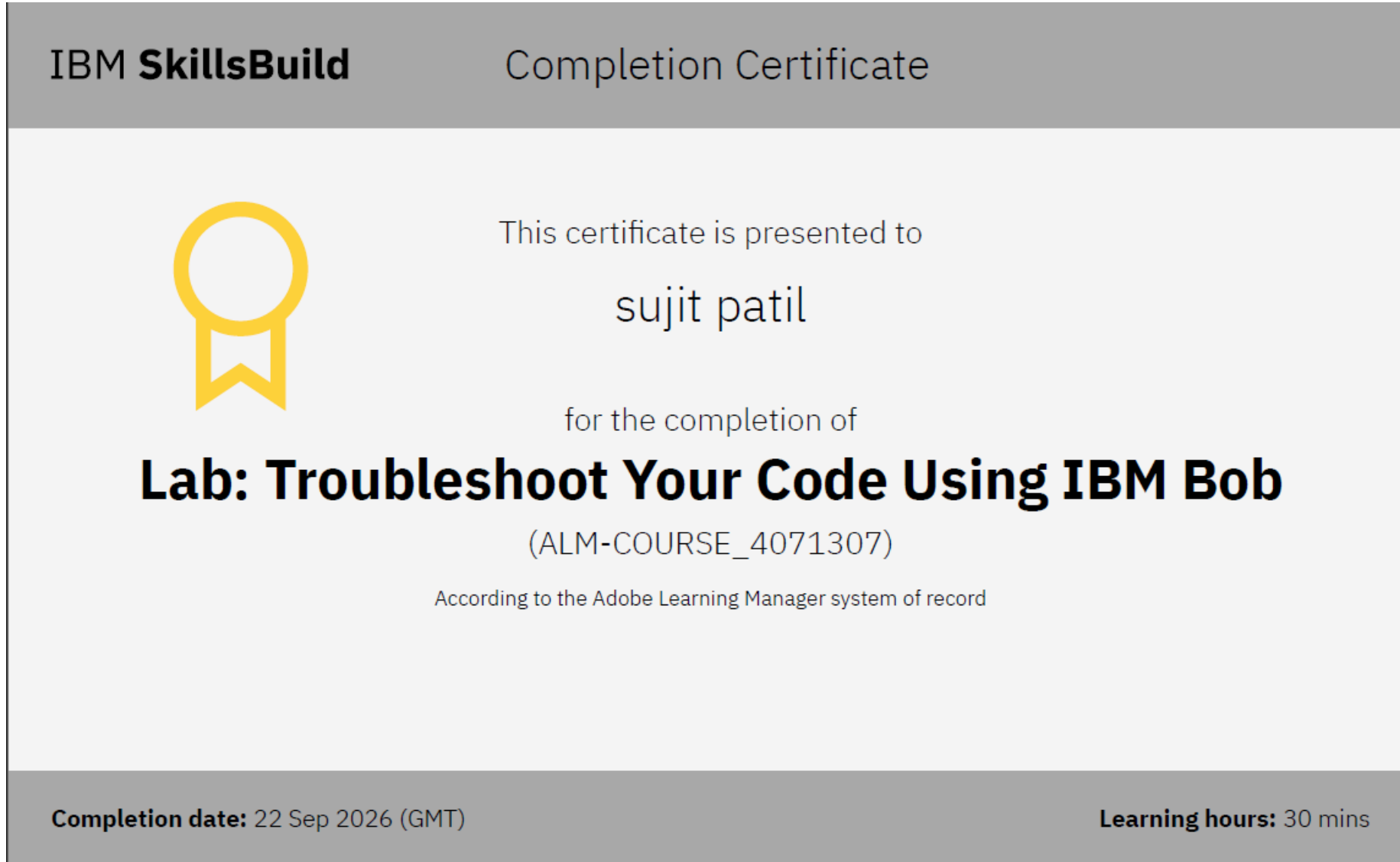
Issued on: Sep 22, 2026
Issued by: IBM SkillsBuild

Verify: <https://www.credly.com/badges/f6668138-f55f-4417-9851-64b37126d50f>



Certificates Earned

IBM BOB Certificate



Thank You!

Thank you for your time and interest.